

A Translation-Invariant Lipschitz Summing Map Whose Pietsch Measure Is Not Haar Measure

Automatic math-discovery run `fa_banach_001`, lane 14

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Result Status

Candidate full counterexample, likely valid, pending expert review. For every $1 \leq p < \infty$, the construction below answers the open problem repeated in Bispo–Macedo–Santos [1] in the negative. It also answers the original 2012 formulation of Botelho–Pellegrino–Rueda–Santos–Seoane–Sepúlveda [2].

1 Source Problem

Let G be a compact Hausdorff topological group, let H be a closed translation-invariant subspace of $C(G)$, let Y be a metric space, and let $u : H \rightarrow Y$ be a translation-invariant Lipschitz p -summing map. The question asks whether normalized Haar measure σ_G must be a Pietsch measure for u [1, p. 2, Introduction, Open Problem].

In the pairwise form used in [1, Section 4], this would mean that there is a finite constant C such that

$$d_Y(u(f), u(g)) \leq C \left(\int_G |f(x) - g(x)|^p d\sigma_G(x) \right)^{1/p} \quad (f, g \in H). \quad (1)$$

The counterexample below rules out even domination with an arbitrary finite constant, and therefore also rules out the sharper convention in which the constant must be $\pi_p^L(u)$.

2 Proof Intuition

The scalar-valued case exposes a simple gap between the hypothesis and the desired conclusion. Every base-point-preserving real-valued Lipschitz function on a metric space is automatically Lipschitz p -summing: after normalization, the function itself is one of the test functions over which the defining supremum is taken.

On $C(G)$, the supremum norm is such a scalar-valued Lipschitz function, and it is invariant under translations of G . Haar domination for this map would force the supremum norm on $C(G)$ to be bounded by its L^p -norm. Continuous peaks on the circle have supremum one and arbitrarily small L^p -norm, giving the contradiction.

3 Counterexample

Theorem 1. *Fix $1 \leq p < \infty$. Let $G = \mathbb{T}$, let $H = C(\mathbb{T}, \mathbb{R})$, let $Y = \mathbb{R}$ with its usual metric, and define*

$$u : H \longrightarrow \mathbb{R}, \quad u(f) = \|f\|_\infty.$$

Then:

1. H is a closed translation-invariant subspace of $C(\mathbb{T})$;
2. u is translation invariant;
3. u is Lipschitz p -summing, in fact $\pi_p^L(u) = 1$;
4. normalized Haar measure on $\mathbb{T}$ is not a Pietsch measure for u .

Consequently, the answer to the source open problem is negative for every finite p .

Proof. The first assertion is immediate. For $\psi \in \mathbb{T}$, the translation $I_\psi(z) = \psi z$ is a bijection of $\mathbb{T}$, and hence

$$u(f \circ I_\psi) = \|f \circ I_\psi\|_\infty = \|f\|_\infty = u(f).$$

Thus u is translation invariant.

We next verify the summing condition directly. The reverse triangle inequality gives

$$|u(f) - u(g)| \leq \|f - g\|_\infty,$$

so $u(0) = 0$ and u belongs to the unit ball of the Lipschitz dual $H^\# = \text{Lip}_0(H, \mathbb{R})$. Therefore, for arbitrary $f_1, \dots, f_m, g_1, \dots, g_m \in H$,

$$\left(\sum_{j=1}^m |u(f_j) - u(g_j)|^p \right)^{1/p} \leq \sup_{v \in B_{H^\#}} \left(\sum_{j=1}^m |v(f_j) - v(g_j)|^p \right)^{1/p}.$$

This is exactly the Farmer–Johnson Lipschitz p -summing inequality, with constant one. Taking $m = 1$, $f_1 \equiv 1$, and $g_1 = 0$ shows that the best constant is one, so $\pi_p^L(u) = 1$.

It remains to disprove Haar domination. Write a point of $\mathbb{T}$ as e^{it} , with $-\pi \leq t \leq \pi$, and let normalized Haar measure be $dt/(2\pi)$. For $n \geq 1$, define the continuous peak

$$f_n(e^{it}) = (1 - n|t|)_+.$$

The endpoint values agree, so this is a well-defined continuous function on the circle. Moreover,

$$\|f_n\|_\infty = 1 \quad \text{and} \quad \int_{\mathbb{T}} |f_n|^p d\sigma_{\mathbb{T}} = \frac{1}{2\pi} \int_{-1/n}^{1/n} (1 - n|t|)^p dt = \frac{1}{\pi n(p+1)}.$$

If normalized Haar measure were a Pietsch measure for u , then applying (1) to $(f_n, 0)$ would give, for some finite C ,

$$1 = |u(f_n) - u(0)| \leq C \left(\frac{1}{\pi n(p+1)} \right)^{1/p} \quad (n \geq 1),$$

which is impossible as $n \rightarrow \infty$. Thus Haar measure is not a Pietsch measure for u . $\square$

4 Robustness and Boundary Checks

Remark 1 (Why the source’s partial theorem is not contradicted). Bispo–Macedo–Santos prove Haar domination for the smaller class of Lipschitz p - B_{H^*} -summing maps [1, Theorem 4.3]. The map above cannot belong to that smaller class: their theorem would then imply (1), which the peaks disprove. Thus the example lands exactly in the gap between ordinary Lipschitz p -summability, tested by the nonlinear Lipschitz dual $H^\#$, and their linear-dual hypothesis.

Remark 2 (No dependence on the optimal constant). The contradiction uses only the nonexistence of any finite domination constant. It is therefore unaffected by differing conventions for the term “Pietsch measure.”

Remark 3 (More general form). The same construction works on any infinite compact group that admits continuous functions of supremum one with Haar L^p -norm tending to zero. The circle was chosen to make this concentration explicit and avoid any auxiliary topological-measure lemma.

5 Novelty Check

The March 2026 source paper still states the question as open and conjectures that its B_{H^*} -based partial theorem is the best possible answer [1, Remark following Theorem 4.3]. Exact-title, arXiv-id, and translation-invariant Lipschitz-summing searches through August 9, 2026 found no later primary source recording this counterexample or another resolution. This is evidence of novelty, not a claim of exhaustive bibliographic proof.

References

- [1] A. Bispo, R. Macedo, and J. Santos, *A new criterion for the normalized Haar measure to be a Pietsch measure*, arXiv:2603.27851 (2026).
- [2] G. Botelho, D. Pellegrino, P. Rueda, J. Santos, and J. B. Seoane-Sepúlveda, *When is the Haar measure a Pietsch measure for nonlinear mappings?*, *Studia Math.* **213** (2012), 275–287; arXiv:1204.5621.
- [3] J. D. Farmer and W. B. Johnson, *Lipschitz p -summing operators*, *Proc. Amer. Math. Soc.* **137** (2009), 2989–2995.